AI-Based Early Mental Health Breakdown Detection from Speech Patterns

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Group 6

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Live Demo: huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis

Why This Matters

280M+ people affected by depression (WHO 2023)

#1 cause of disability worldwide

Diagnosis = subjective PHQ-9 / HAMD-17 questionnaires

→ requires trained clinicians, time-consuming, self-report bias

Treatment gap > 50% even in high-income countries

SPEECH = OBJECTIVE BIOMARKER

Depressed speech has measurable differences:

- ↓ F0 variability (flat prosody)
- ↑ Pause duration
- ↓ Speech rate
- ↑ Jitter / Shimmer
- ↓ NAQ (weak vocal fold closure)

AUC 0.70–0.88 across 28 studies (Trevisan 2019)

CAN WE BUILD A SCALABLE, NON-INVASIVE SCREENING TOOL USING JUST SPEECH?

No clinician needed • No self-report bias • Works on any phone recording

What's Missing in Current Methods

Single-Dataset Evaluation

Most papers test on ONE dataset only → no generalization evidence

We evaluate on 4 independent benchmarks (RAVDESS, DAIC-WOZ, MODMA, SWELL)

Wrong Metrics

Accuracy reported despite 2.57:1 class imbalance → a majority predictor gets 72%

We use Macro-F1 + Sensitivity + AUPRC as primary metrics

No Public Deployment

Models stay in notebooks → zero clinical accessibility

We deploy a live app with RAG clinical explainer on HF Spaces

Feature Engineering Burden

Hand-tuned MFCCs/jitter per dataset → doesn't scale

Frozen Whisper embeddings = zero feature engineering, 680K hrs pretraining

Two Pipelines — Best of Both Worlds

PIPELINE 1 — Handcrafted + Classical ML

Raw Audio

↓ librosa feature extraction

↓ MFCCs + F0 + Jitter + Shimmer + Formants

↓ COVAREP / NLP features (DAIC-WOZ)

↓ **338 – 2,231 features**

↓ RobustScaler

↓ XGBoost / SVM / MLP

Best for: interpretability, cross-lingual, non-speech signals (HRV/EDA)

PIPELINE 2 — Frozen Whisper + ML Head

Raw Audio

↓ Resample 16 kHz, pad/trim 30s

↓ 80-bin log mel-spectrogram

↓ **Whisper encoder (FROZEN, 74-244M params)**

↓ Mean + Std temporal pooling

↓ **512-D or 1536-D embedding**

↓ XGBoost / SVM / Fusion MLP

Best for: English speech, small datasets (frozen = no overfitting on N=107)

4 Benchmarks — One Framework

RAVDESS

8-class Emotion

1,440 files
24 actors (12M/12F)
Actor-level splits

Balanced

DAIC-WOZ

Binary Depression

189 interviews
PHQ-8 ≥ 10
Participant splits

2.57 : 1

MODMA

MDD (Mandarin)

52 subjects
23 MDD + 29 HC
Subject splits

Cross-lingual

SWELL-KW

3-class Stress

25 workers
HRV + EDA
256K windows

Non-speech

All splits are participant/actor-level ⟶ prevents speaker identity leakage (inflates metrics 5-15%)

Pipeline Architecture — Visual Overview

Raw Audio → Mel-Spec 80 bins → Whisper Encoder (FROZEN) → Mean+Std Pooling → 512-D or 1536-D → XGBoost or MLP → Prediction

XGBoost Config

max_depth=2-3 | lr=0.05-0.10 | subsample=0.7-0.8

reg_alpha=0.1-0.5 | reg_lambda=1.5-2.0

scale_pos_weight=2.567 (DAIC-WOZ imbalance)

eval_metric=aucpr (sensitive under imbalance)

SVM Config (MODMA best)

Kernel=RBF | C=100 | gamma='scale'

CalibratedClassifierCV (isotonic, 5-fold)

Why SVM: N=36 train → margin-maximization

provides strong implicit regularization

The Numbers That Matter

97.4%

Macro-F1 • RAVDESS Emotion

+48.8 pp over classical baseline

(Whisper + XGBoost)

60.7%

Macro-F1 • DAIC-WOZ Depression

Sensitivity = 85.7% (6/7 caught)

(Whisper + Fusion MLP)

87.3%

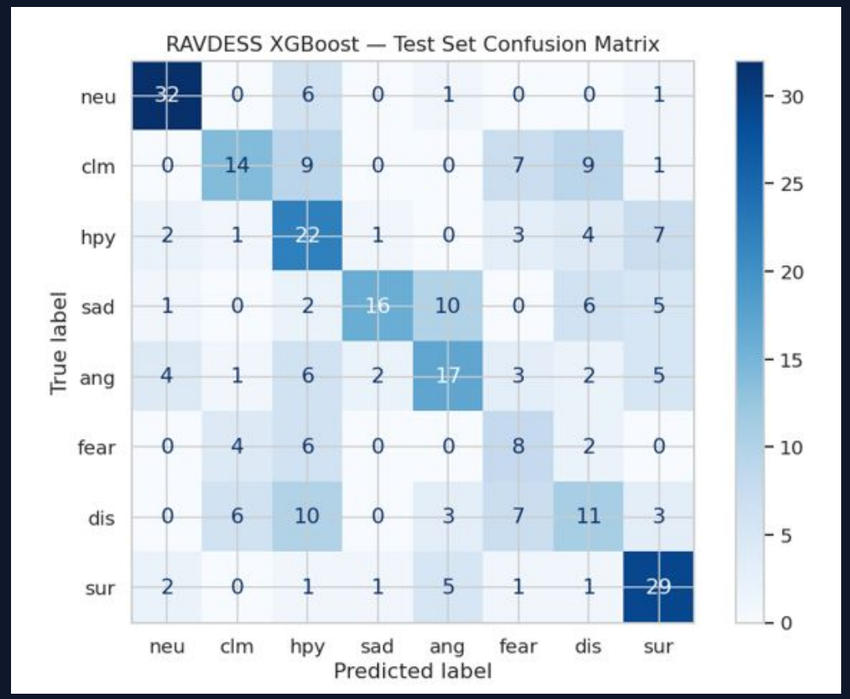
Macro-F1 • MODMA (Mandarin)

Classical SVM beats Whisper here

(SVM RBF C=100)

Cross-Task Summary

Task	Baseline (P1)	Best (P2)	Delta	Verdict
RAVDESS (8-class)	F1 = 0.486	F1 = 0.974	+48.8 pp	✅ Whisper transformative
DAIC-WOZ (depression)	F1 = 0.507	F1 = 0.607	+10.0 pp	✅ Clinical improvement
MODMA (Mandarin MDD)	F1 = 0.873	F1 = 0.750	-12.3 pp	❌ Cross-lingual failure
SWELL (3-class stress)	F1 = 0.508	—	No gain	⚠️ Non-speech signals



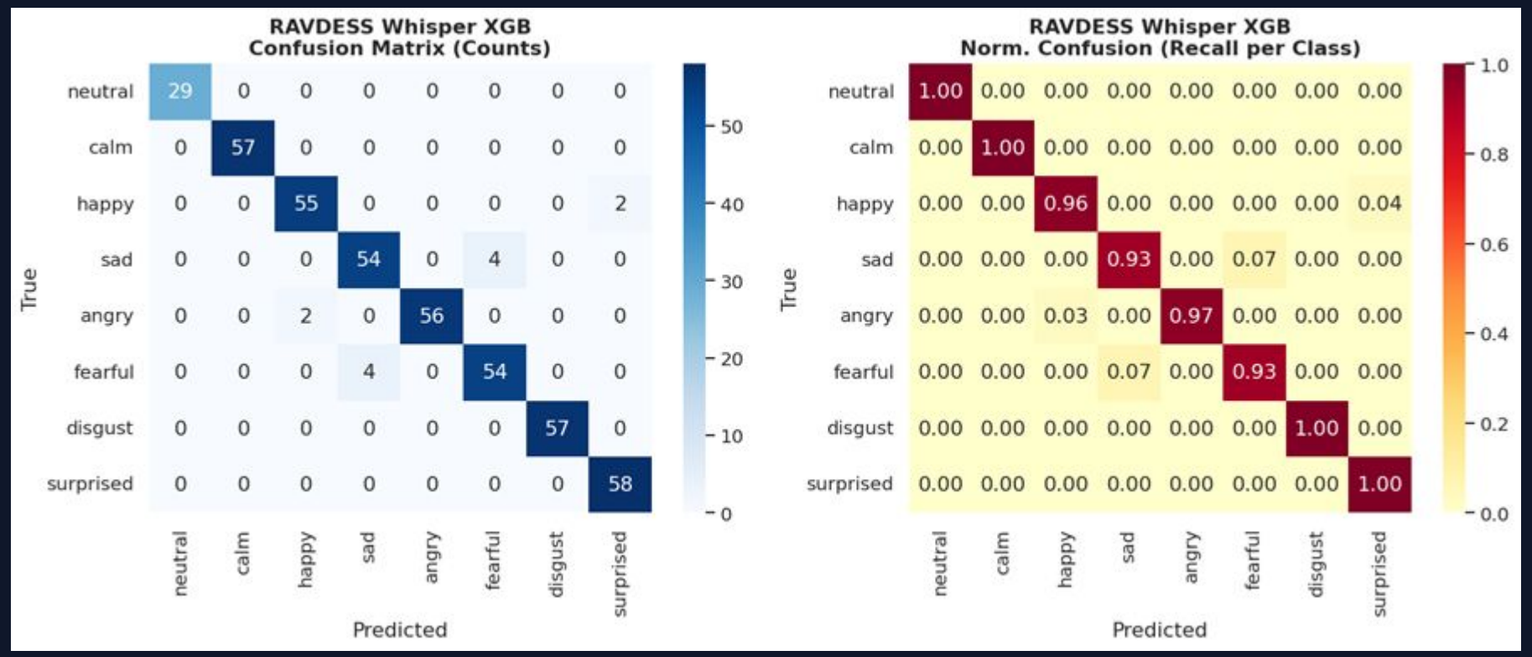


Figure 5.1 -- RAVDESS XGBoost MFCC baseline confusion matrix. Significant off-diagonal mass in acoustically similar pairs: Calm<->Neutral, Sad<->Fearful

Figure 5.2 -- RAVDESS Whisper+XGBoost (Pipeline 2) confusion matrix and per-class error breakdown. Near-diagonal concentration confirms 97.4% macro-F1. Residual errors: Calm/Neutral boundary only.

The Threshold Story

THE PROBLEM

Default threshold = 0.50

At 2.57:1 class imbalance, the model's posterior P(depressed|x) is shifted toward the majority class (non-depressed).

THE FIX

Sweep threshold on validation set

Optimal threshold = 0.45 (just a 0.05 shift!)

Result: Sensitivity = 0.857

→ Catches 6 of 7 depressed cases

THRESHOLD SWEEP RESULTS

Threshold	Macro-F1	Sensitivity	Specificity	Clinical Verdict
0.50	0.480	0.350	0.900+	❌ Misses most depressed cases
0.45	**0.607**	**0.857**	**0.575**	✅ **SELECTED — best F1 + clinical**
0.35	0.540	0.929	0.425	⚠️ Too many false positives
0.20	0.510	1.000	0.275	⚠️ Over-screens everyone

💡 **This 0.05 threshold shift produced MORE improvement than any model architecture change**

Figure 5.3 -- DAIC-WOZ Fusion MLP (left) and MODMA SVM (right) confusion matrices. DAIC-WOZ shows false-negative bias from 2.57:1 imbalance; MODMA shows near-perfect separation.

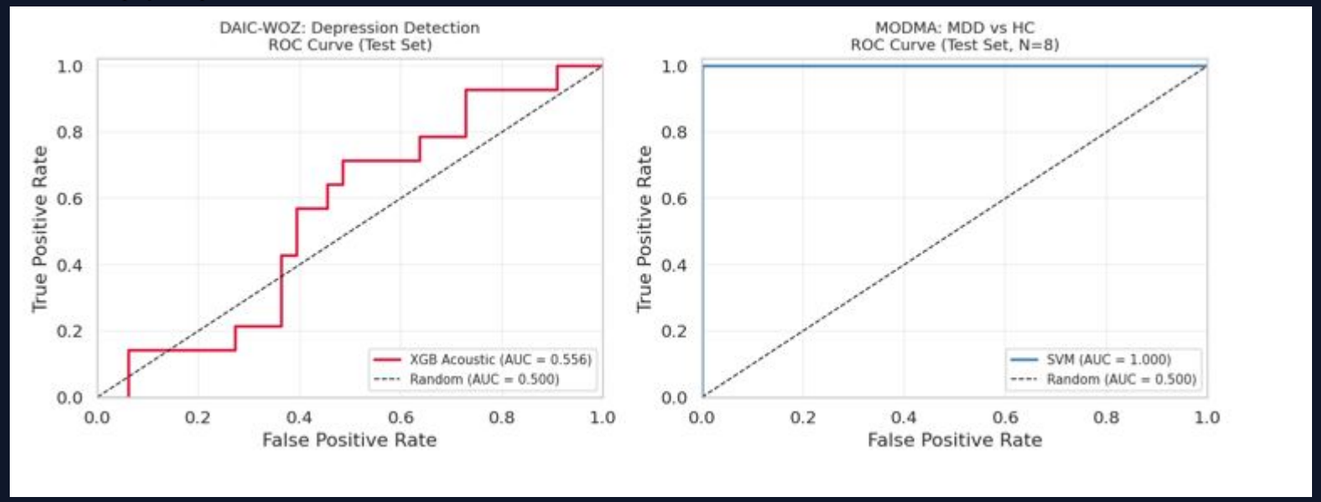
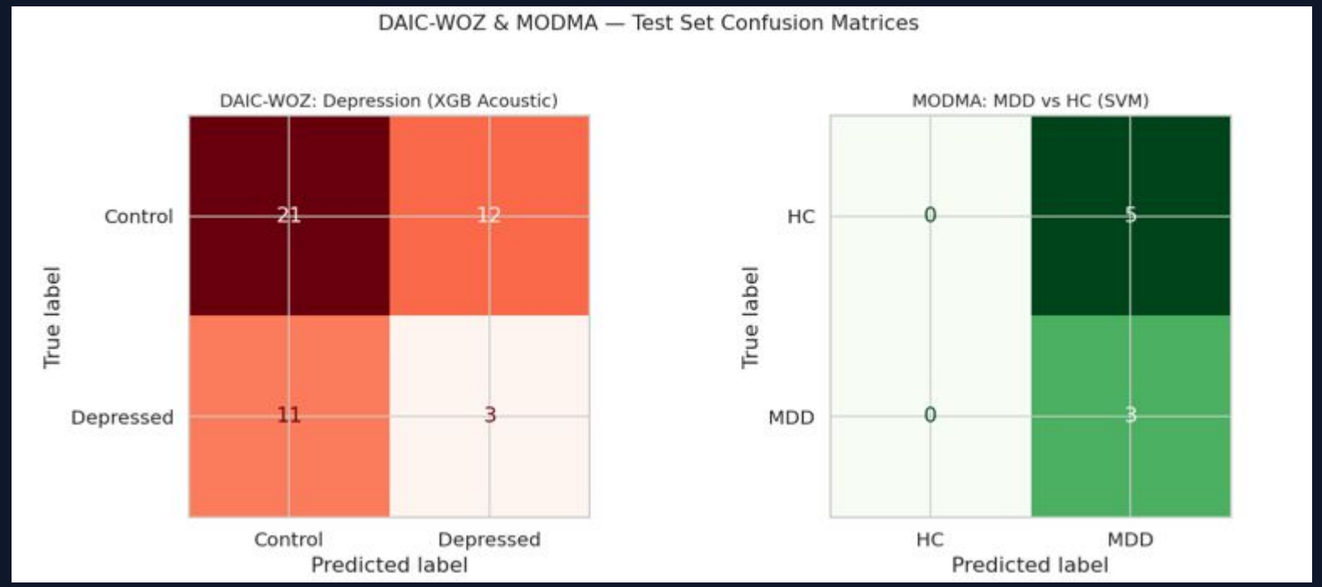


Figure 5.3 -- ROC curves for DAIC-WOZ (Fusion MLP) and MODMA (SVM). DAIC-WOZ curve shows sensitivity/specificity tradeoff across thresholds.

Figure 5.4 -- DAIC-WOZ Fusion MLP (left) and MODMA SVM (right) confusion matrices. DAIC-WOZ shows false-negative bias from 2.57:1 imbalance; MODMA shows near-perfect separation.



Honest Limitations

Small Sample Sizes

N=189 (DAIC-WOZ), N=52 (MODMA)
→ Power analysis suggests N>500 for generalization

Acted ≠ Real

97.4% on RAVDESS actors ≠ clinical speech
Actors produce exaggerated, unambiguous emotions

Cross-Lingual Failure

Whisper (English) loses -12.3pp on Mandarin
Tonal phonology confounds affective prosody

PHQ-8 Regression

RMSE = 6.515 on 0-24 scale
→ Cannot distinguish mild from moderate depression

Audio-Only

Missing text + facial + clinical history
Visual branch achieved only F1=0.40 (EDA)

Not a Medical Device

No FDA/CE approval, no uncertainty quantification
Research prototype only — not for diagnosis

Live on Hugging Face Spaces

Tab 1

Emotion Recognition

Whisper-small (1536D)
+ XGBoost

F1 = 0.974

Tab 2

Depression Screening

Whisper-small (1536D)
+ SVM RBF

F1 = 0.750

Tab 3

PHQ-8 Self-Screener

Rule-based scoring
(≥10 = depressed)

Validated

Tab 4

AI Clinical Explainer

RAG + FAISS + Groq
LLaMA3-70B

temp=0.3

Tab 5

About & Details

Model cards,
disclaimer, links

Static

RAG Pipeline (Tab 4)

Audio ⟶ ML Predict ⟶ Build Query ⟶ FAISS Retrieval (15 curated clinical chunks, all-MiniLM-L6-v2)

⟶ Top-4 chunks ⟶ Groq LLaMA3-70B (temp=0.3) ⟶ 4-paragraph clinical report

Why RAG? Grounding prevents hallucination on medical topics • Curated KB = auditable + citable

Three Things to Remember

01 Foundation Models ≠ Always Better

English: +48.8 pp (RAVDESS), +10 pp (DAIC-WOZ)
Mandarin: -12.3 pp

Pretraining domain alignment > model scale

02 Threshold > Architecture

A 0.05 shift (0.50→0.45) improved sensitivity from 35% to 85.7%

Post-hoc threshold tuning is the single most impactful intervention

03 Freeze for Small Clinical Data

N=107 (DAIC-WOZ): fine-tuning 244M params = overfitting by epoch 2-3

Frozen encoder + lightweight head = optimal bias-variance tradeoff

Key References

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Thank You

97.4%	85.7%	4	86	LIVE
Emotion F1	Sensitivity	Benchmarks	Experiments	HF Spaces

huggingface.co/spaces/PANKAJ-MOHAN/mental-health-speech-analysis

github.com/21f3002286/Group-6-DS-and-AI-Lab-Project

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Q & A